

# HVLED101 Single Stage Flyback Converter Design Guide

## 1. Design Specifications

|                                        |                                                         |                                    |                                                                |
|----------------------------------------|---------------------------------------------------------|------------------------------------|----------------------------------------------------------------|
| - Input voltage range:                 | $V_{AC\_min} := 90 \text{ V}$                           | $V_{AC\_max} := 265 \text{ V}$     |                                                                |
| - Minimum Input frequency:             | $f_{line\_min} := 45 \text{ Hz}$                        |                                    |                                                                |
| - Secondary #1 Output Voltage/Current: | $V_{out1} := 60 \text{ V}$                              | $I_{out1\_max} := 0.833 \text{ A}$ | $P_{out1\_max} := V_{out1} \cdot I_{out1\_max} = 50 \text{ W}$ |
| - Aux Output Voltage/Current:          | $V_{aux} := 41 \text{ V}$                               | $I_{aux\_max} := 25 \text{ mA}$    | $P_{aux\_max} := V_{aux} \cdot I_{aux\_max} = 1.025 \text{ W}$ |
| - Maximum 2fL output ripple:           | $\Delta V_{out1} := V_{out1} \cdot 5 \% = 3 \text{ V}$  |                                    |                                                                |
|                                        | $\Delta V_{aux} := V_{aux} \cdot 10 \% = 4.1 \text{ V}$ |                                    |                                                                |

## 2. Pre-design Choices

|                                                      |                                                       |
|------------------------------------------------------|-------------------------------------------------------|
| - Minimum switching frequency:                       | $f_{sw\_min} := 70 \text{ kHz}$                       |
| - Primary-to-secondary output #1 turns ratio Np/Ns1: | $n := 2.25$                                           |
| - Leakage inductance Spike:                          | $V_{spike} := 120 \text{ V}$                          |
| - Expected efficiency:                               | $\eta := 90 \%$                                       |
| - Secondary Rectifier Forward Voltage:               | $V_f := 0.7 \text{ V}$                                |
| - Reflected voltage:                                 | $V_R := n \cdot (V_{out1} + V_f) = 136.575 \text{ V}$ |
| - Input Bridge Diode Forward Voltage:                | $V_{rect} := 1.1 \text{ V}$                           |

### 3. Preliminary Calculations

- Minimum Input Peak Voltage:

$$V_{pk\_min} := V_{AC\_min} \cdot \sqrt{2} - 2 \cdot V_{rect} = 125.0792 \text{ V}$$

- Maximum Input Peak Voltage:

$$V_{pk\_max} := V_{AC\_max} \cdot \sqrt{2} - 2 \cdot V_{rect} = 372.5666 \text{ V}$$

- Maximum Output Power:

$$P_{out\_max} := P_{out1\_max} + P_{aux\_max} = 51.005 \text{ W}$$

- Maximum Input Power:

$$P_{in\_max} := \frac{P_{out\_max}}{\eta} = 56.6722 \text{ W}$$

- Peak-to-reflected Voltage Ratio:

$$K_v := \frac{V_{pk\_min}}{V_R} = 0.9158$$

- Characteristic functions value:

$$F1 := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))}{1 + K_v \cdot \sin(\theta)} d\theta = 0.3766$$

$$F2 := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_v \cdot \sin(\theta)} d\theta = 0.2839$$

$$F3 := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^3}{1 + K_v \cdot \sin(\theta)} d\theta = 0.2359$$

### 4. Operating Conditions

- Peak Primary Current:

$$I_{pri\_pk} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_min} \cdot F2} = 3.19 \text{ A}$$

- RMS Primary Current:

$$I_{pri\_rms} := I_{pri\_pk} \cdot \sqrt{\frac{F2}{3}} = 0.98 \text{ A}$$

- Peak Secondary Vout #1 Current:

$$I_{sec1\_pk} := 2 \cdot \frac{I_{out1\_max}}{K_v \cdot F2} = 6.41 \text{ A}$$

- RMS Secondary Vout #1 Current:

$$I_{sec1\_rms} := I_{sec1\_pk} \cdot \sqrt{K_v \cdot \frac{F3}{3}} = 1.72 \text{ A}$$

## 5. Transformer

### - Primary inductance:

$$L_p := \frac{V_{pk\_min}}{(1 + K_v) \cdot f_{sw\_min} \cdot I_{pri\_pk}} = 292.2265 \mu\text{H}$$

$$L_{p\_min} := \frac{(4.3 \mu\text{s}) \cdot V_R}{(0.93) \cdot I_{pri\_pk}} = 197.85 \mu\text{H}$$

### - Minimum area product calculation:

$$A_{p\_min} := \left( 460 \cdot \frac{P_{in\_max}}{f_{sw\_min} \cdot (1 + K_v) \cdot \sqrt{F2}} \frac{\text{Hz}}{\text{W}} \right)^{1.316} \text{cm}^4 = 0.27 \text{cm}^4$$

### - Effective magnetic path length:

$$l_e := 46.3 \text{ mm}$$

### - Minimum core cross section:

$$A_e := 119 \text{ mm}^2$$

### - Effective magnetic cross section:

$$A_{min} := 58.1 \text{ mm}^2$$

### - Winding cross section:

$$A_N := 60.4 \text{ mm}^2$$

### - Area product calculation:

$$A_p := A_{min} \cdot A_N = 0.35 \text{ cm}^4$$

### - Air Gap:

$$g := 2 \cdot (0.35) \text{ mm} = 0.7 \text{ mm}$$

### - Inductance factor:

$$A_L := \left( \frac{g}{\text{mm}} \right)^{-0.8106} \cdot 173.76 \text{ nH} = 232.01 \text{ nH}$$

### - Number of Primary Turns:

$$N_p := \sqrt{\frac{L_p}{A_L}} = 35.5$$

### - Number of Secondary #1 Turns:

$$N_{s1} := \frac{N_p}{n} = 15.8$$

### - Number of Auxiliary Turns:

$$N_{aux} := \frac{V_{aux}}{V_{out1}} \cdot N_{s1} = 10.8$$

### - Maximum Flux Density Swing:

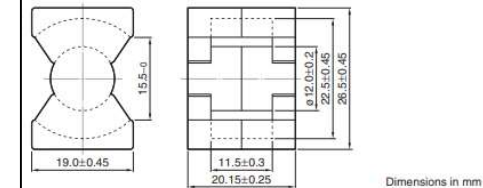
$$B_{max} := 0.3 \text{ T}$$

### - Transformer Saturation Current:

$$I_{sat} := \frac{N_p \cdot B_{max} \cdot A_e}{L_p} = 4.3356 \text{ A}$$

## Mn-Zn PQ series Part No.: PC90PQ26/20Z-12

### SHAPES AND DIMENSIONS

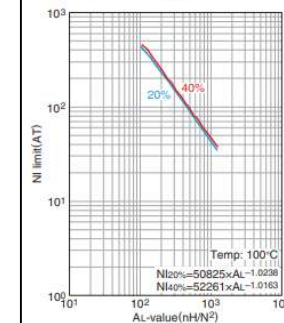


| Effective parameter                      |                                                          |                                                                        |                                                               |                                                                           |                                                                                       |                                                                               |                   | Electrical characteristics                          |                                                   |
|------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------|-----------------------------------------------------|---------------------------------------------------|
| Core factor<br>C1<br>(mm <sup>-1</sup> ) | Effective magnetic path length<br>l <sub>e</sub><br>(mm) | Effective cross-sectional area<br>A <sub>e</sub><br>(mm <sup>2</sup> ) | Effective core volume<br>V <sub>e</sub><br>(mm <sup>3</sup> ) | Cross-sectional center pole area<br>A <sub>cp</sub><br>(mm <sup>2</sup> ) | Minimum cross-sectional center pole area<br>A <sub>cp min</sub><br>(mm <sup>2</sup> ) | Cross-sectional winding area of core<br>A <sub>ow</sub><br>(mm <sup>2</sup> ) | Weight<br>(g/set) | AL-value *<br>(nH/N <sup>2</sup> )<br>1kHz<br>0.5mA | Core loss<br>(WJmax.<br>100kHz<br>200mT<br>100°C) |
| 0.391                                    | 46.3                                                     | 119                                                                    | 5490                                                          | 113                                                                       | 109                                                                                   | 60.4                                                                          | 31                | 5500±25%                                            | 2.45                                              |

\* Coil : ø0.35 2UEW 100Ts

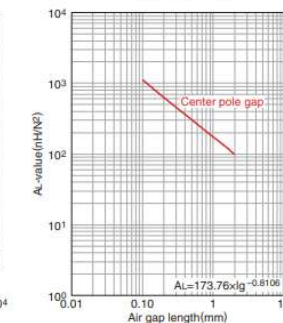
○ Calculated output power (forward converter mode): 145W

### NI limit vs. AL-value (Typ.)



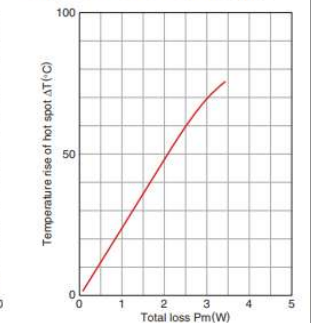
The 20% and 40% graph shows when a 20% and 40% drop from the initial AL-value has been made due to the DC superimposition.

### AL-value vs. Air gap length (Typ.)

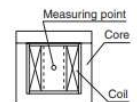


Measuring conditions  
• Coil : ø0.35 2UEW 100Ts  
• Frequency : 1kHz  
• Current level : 0.5mA  
• Ambient temperature : 25°C

### Temperature rise vs. Total loss (Typ.)



Measuring conditions  
• Room space: approx. 400x300x300cm  
• Ambient temperature : 25°C  
• Humidity: 45%(%)RH.



- Current Density:

$$J := 7 \frac{\text{A}}{\text{mm}^2}$$

- Primary Winding Wire Cross Sectional Area :

$$W_{pri\_area} := \frac{I_{pri\_rms}}{J} = 0.1403 \text{ mm}^2$$

- Primary Winding Wire Equivalent Minimum Diameter :

$$\phi_{pri} := 2 \cdot \left( \sqrt{\frac{W_{pri\_area}}{\pi}} \right) = 0.4226 \text{ mm}$$

- Secondary #1 Winding Wire Cross Sectional Area :

$$W_{sec1\_area} := \frac{I_{sec1\_rms}}{J} = 0.2456 \text{ mm}^2$$

- Secondary #1 Winding Wire Equivalent Minimum Diameter :

$$\phi_{sec1} := 2 \cdot \left( \sqrt{\frac{W_{sec1\_area}}{\pi}} \right) = 0.5592 \text{ mm}$$

- The skin depth at 100 °C in millimeters:

$$V_{rms\_264VAC} := 264 \text{ V} \quad L_p = 292.23 \text{ }\mu\text{H}$$

$$V_{pk\_264VAC} := V_{rms\_264VAC} \cdot \sqrt{2} - 2 \cdot V_{rect} = 371 \text{ V}$$

$$K_{v\_264VAC} := \frac{V_{pk\_264VAC}}{V_R} = 2.72$$

$$F2_{264VAC} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_{v\_264VAC} \cdot \sin(\theta)} d\theta = 0.16$$

$$I_{pri\_pk\_264VAC} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_264VAC} \cdot F2_{264VAC}} = 1.97 \text{ A}$$

$$f_{sw\_max} := \frac{V_{pk\_264VAC}}{(1 + K_{v\_264VAC}) \cdot L_p \cdot I_{pri\_pk\_264VAC}} = 173.84 \text{ kHz}$$

$$\delta := \frac{76}{\sqrt{\frac{f_{sw\_max}}{\text{Hz}}}} \text{ mm} = 0.1823 \text{ mm}$$

All Wire Diameter should be less than 0.3646 mm

- Setting Primary Winding Wire :

$$\phi_{pri} := 0.3 \text{ mm} \quad strands := 2$$

$$W_{pri\_area} := strands \cdot \pi \cdot \left( \frac{\phi_{pri}}{2} \right)^2 = 0.141 \text{ mm}^2$$

- Setting Secondary #1 Winding Wire :

$$\phi_{sec1} := 0.4 \text{ mm} \quad strands := 2$$

$$W_{sec1\_area} := strands \cdot \pi \cdot \left( \frac{\phi_{sec1}}{2} \right)^2 = 0.251 \text{ mm}^2$$

**6. MOSFET selection****- Maximum Drain Voltage:**

$$V_{DS\_max} := V_{pk\_max} + V_R + V_{spike} = 629.1 \text{ V}$$

**7. Secondary diode selection****- Maximum reverse voltage for Output #1 Diode:**

$$V_{REV1\_max} := \frac{(V_{pk\_max} + V_{spike})}{n} + V_{out1} = 278.9 \text{ V}$$

**8. Output Capacitor Selection****- The minimum capacitance value that meets the specification on the 100/120 Hz ripple:**

$$\vec{H2} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2 \cdot \cos(2 \cdot \theta)}{1 + K_v \cdot \sin(\theta)} d\theta = -0.1277$$

$$C_{out1\_min} := \frac{1}{\pi \cdot f_{line\_min}} \cdot \frac{|\vec{H2}|}{F2} \cdot \frac{I_{out1\_max}}{\Delta V_{out1}} = 883 \text{ }\mu\text{F}$$

**- Vout #1 Hold-up time 20ms Min Capacitance:**

$$t_{holdup} := 20 \text{ ms} \quad V_{out1\_deep} := V_{out1} \cdot 0.7 = 42 \text{ V}$$

$$E_{Holdup\_energy} := P_{out1\_max} \cdot t_{holdup} = 0.9996 \text{ J}$$

$$C_{out1\_Holdup} := 2 \cdot \frac{E_{Holdup\_energy}}{\left( (V_{out1} - \Delta V_{out1})^2 - (V_{out1\_deep})^2 \right)} = 1346 \text{ }\mu\text{F}$$

## 9. Design Review and Summary

**- Input 90VAC:**

$$V_{rms\_90VAC} := 90 \text{ V} \quad L_p = 292.23 \text{ }\mu\text{H}$$

$$V_{pk\_90VAC} := V_{rms\_90VAC} \cdot \sqrt{2} - 2 \cdot V_{rect} = 125 \text{ V}$$

$$K_{v\_90VAC} := \frac{V_{pk\_90VAC}}{V_R} = 0.92$$

$$F2_{90VAC} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_{v\_90VAC} \cdot \sin(\theta)} d\theta = 0.28$$

$$I_{pri\_pk\_90VAC} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_90VAC} \cdot F2_{90VAC}} = 3.19 \text{ A}$$

$$f_{sw\_90VAC} := \frac{V_{pk\_90VAC}}{(1 + K_{v\_90VAC}) \cdot L_p \cdot I_{pri\_pk\_90VAC}} = 70 \text{ kHz}$$

$$D_{max\_90VAC} := \text{solve} \left( V_{pk\_90VAC} \cdot \frac{1}{n} \cdot \frac{D}{1-D} - V_{out1}, D \right) = 0.52$$

$$T_{sw\_90VAC} := \frac{1}{f_{sw\_90VAC}} = 14.2857 \text{ }\mu\text{s}$$

$$t_{on\_max\_90VAC} := T_{sw\_90VAC} \cdot D_{max\_90VAC} = 7.42 \text{ }\mu\text{s}$$

$$t_{off\_min\_90VAC} := T_{sw\_90VAC} \cdot (1 - D_{max\_90VAC}) = 6.871 \text{ }\mu\text{s}$$

$$T_{demag\_90VAC} := L_p \cdot \frac{I_{pri\_pk\_90VAC}}{V_R} = 6.83 \text{ }\mu\text{s}$$

**- Input 110VAC:**

$$V_{rms\_110VAC} := 110 \text{ V} \quad L_p = 292.23 \text{ }\mu\text{H}$$

$$V_{pk\_110VAC} := V_{rms\_110VAC} \cdot \sqrt{2} - 2 \cdot V_{rect} = 153 \text{ V}$$

$$K_{v\_110VAC} := \frac{V_{pk\_110VAC}}{V_R} = 1.12$$

$$F2_{110VAC} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_{v\_110VAC} \cdot \sin(\theta)} d\theta = 0.26$$

$$I_{pri\_pk\_110VAC} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_110VAC} \cdot F2_{110VAC}} = 2.85 \text{ A}$$

$$f_{sw\_110VAC} := \frac{V_{pk\_110VAC}}{(1 + K_{v\_110VAC}) \cdot L_p \cdot I_{pri\_pk\_110VAC}} = 86.64 \text{ kHz}$$

$$D_{max\_110VAC} := \text{solve} \left( V_{pk\_110VAC} \cdot \frac{1}{n} \cdot \frac{D}{1-D} - V_{out1}, D \right) = 0.47$$

$$T_{sw\_110VAC} := \frac{1}{f_{sw\_110VAC}} = 11.54 \text{ }\mu\text{s}$$

$$t_{on\_max\_110VAC} := T_{sw\_110VAC} \cdot D_{max\_110VAC} = 5.4 \text{ }\mu\text{s}$$

$$t_{off\_min\_110VAC} := T_{sw\_110VAC} \cdot (1 - D_{max\_110VAC}) = 6.14 \text{ }\mu\text{s}$$

$$T_{demag\_110VAC} := L_p \cdot \frac{I_{pri\_pk\_110VAC}}{V_R} = 6.1 \text{ }\mu\text{s}$$

**- Input 220VAC:**

$$V_{rms\_220VAC} := 220 \text{ V} \quad L_p = 292.23 \text{ }\mu\text{H}$$

$$V_{pk\_220VAC} := V_{rms\_220VAC} \cdot \sqrt{2} - 2 \cdot V_{rect} = 309 \text{ V}$$

$$K_{v\_220VAC} := \frac{V_{pk\_220VAC}}{V_R} = 2.26$$

$$F2_{220VAC} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_{v\_220VAC} \cdot \sin(\theta)} d\theta = 0.18$$

$$I_{pri\_pk\_220VAC} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_220VAC} \cdot F2_{220VAC}} = 2.09 \text{ A}$$

$$f_{sw\_220VAC} := \frac{V_{pk\_220VAC}}{(1 + K_{v\_220VAC}) \cdot L_p \cdot I_{pri\_pk\_220VAC}} = 154.86 \text{ kHz}$$

$$D_{max\_220VAC} := \text{solve} \left( V_{pk\_220VAC} \cdot \frac{1}{n} \cdot \frac{D}{1-D} - V_{out1}, D \right) = 0.3$$

$$T_{sw\_220VAC} := \frac{1}{f_{sw\_220VAC}} = 6.4574 \text{ }\mu\text{s}$$

$$t_{on\_max\_220VAC} := T_{sw\_220VAC} \cdot D_{max\_220VAC} = 1.96 \text{ }\mu\text{s}$$

$$t_{off\_min\_220VAC} := T_{sw\_220VAC} \cdot (1 - D_{max\_220VAC}) = 4.494 \text{ }\mu\text{s}$$

$$T_{demag\_220VAC} := L_p \cdot \frac{I_{pri\_pk\_220VAC}}{V_R} = 4.48 \text{ }\mu\text{s}$$

**- Input 264VAC:**

$$V_{rms\_264VAC} := 264 \text{ V} \quad L_p = 292.23 \text{ }\mu\text{H}$$

$$V_{pk\_264VAC} := V_{rms\_264VAC} \cdot \sqrt{2} - 2 \cdot V_{rect} = 371 \text{ V}$$

$$K_{v\_264VAC} := \frac{V_{pk\_264VAC}}{V_R} = 2.72$$

$$F2_{264VAC} := \frac{1}{\pi} \cdot \int_0^{\pi} \frac{(\sin(\theta))^2}{1 + K_{v\_264VAC} \cdot \sin(\theta)} d\theta = 0.16$$

$$I_{pri\_pk\_264VAC} := 2 \cdot \frac{P_{in\_max}}{V_{pk\_264VAC} \cdot F2_{264VAC}} = 1.97 \text{ A}$$

$$f_{sw\_264VAC} := \frac{V_{pk\_264VAC}}{(1 + K_{v\_264VAC}) \cdot L_p \cdot I_{pri\_pk\_264VAC}} = 173.84 \text{ kHz}$$

$$D_{max\_264VAC} := \text{solve} \left( V_{pk\_264VAC} \cdot \frac{1}{n} \cdot \frac{D}{1-D} - V_{out1}, D \right) = 0.27$$

$$T_{sw\_264VAC} := \frac{1}{f_{sw\_264VAC}} = 5.75 \text{ }\mu\text{s}$$

$$t_{on\_max\_264VAC} := T_{sw\_264VAC} \cdot D_{max\_264VAC} = 1.53 \text{ }\mu\text{s}$$

$$t_{off\_min\_264VAC} := T_{sw\_264VAC} \cdot (1 - D_{max\_264VAC}) = 4.22 \text{ }\mu\text{s}$$

$$T_{demag\_264VAC} := L_p \cdot \frac{I_{pri\_pk\_264VAC}}{V_R} = 4.21 \text{ }\mu\text{s}$$